

SIMATS ENGINEERING

ASSESSMENT TOOL 1: AI Model Design Exercise

COURSE NAME: **ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01**
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.*(BTL 3)

Assessment Tool Weightage: 40 %

Total Marks: 100

Time: 90 Mins

No. of Questions: 1

Annexure A AI Model Design Exercise

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Dataset Selection and Data Understanding	10%	
3. Data Preprocessing and Feature Engineering	10%	
4. AI Technique Selection and Justification	10%	
5. Model Design / Architecture	10%	
6. Algorithm / Pseudocode Development	5%	
7. Model Implementation and GitHub Repository	15%	
8. Experimental Design and Results	10%	
9. Performance Evaluation and Metrics	10%	
10. Result Analysis and Interpretation	5%	
11. Limitations and Future Enhancements	5%	
12. Documentation and Technical Presentation	10%	

Total	100%	
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Signature of the Faculty

Total Marks Awarded

S.No.	Scenario / Problem Statement	AI Model / Technique to be Developed	Expected Development Outcome
1	Smart Loan Approval: A bank wants to automatically determine whether a customer is eligible for a loan using income, credit score, employment status, loan amount, and repayment history.	Decision Tree	Design and implement a decision-tree-based AI model and determine the most suitable attribute-selection measure such as Information Gain, Gain Ratio, or Gini Index.

SMART LOAN APPROVAL PREDICTION USING DECISION TREE

1. Problem Statement

A bank needs to automate the preliminary evaluation of loan applications. The system uses income, credit score, employment status, loan amount, and repayment history to predict whether a customer is eligible for a loan. The AI task is a classification problem with two possible outputs: Loan Approved or Loan Rejected.

2. Objectives

- To develop a Decision Tree model for loan eligibility prediction.
- To identify important factors affecting loan approval.
- To preprocess the applicant data for model training.
- To evaluate the model using suitable classification metrics.
- To provide an interpretable loan approval prediction.

3. Dataset Description

The dataset contains customer information required for loan eligibility prediction.

Attribute Description Type

Income	Applicant's income	Numerical
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Credit Score	Applicant's creditworthiness score	Numerical
Employment Status	Applicant's employment condition	Categorical
Loan Amount	Requested loan amount	Numerical
Repayment History	Previous repayment record	Categorical
Loan Status	Approval decision	Target

Target variable: Loan Status

Classes: Approved / Rejected

4. Data Preprocessing

The following preprocessing steps are applied:

1. Remove duplicate or invalid records.
2. Handle missing values appropriately.
3. Encode categorical attributes into numerical values.
4. Separate input features from the target variable.
5. Split the dataset into training and testing sets.
6. Check the processed data before model training.

5. AI TECHNIQUE AND JUSTIFICATION

Selected Technique: Decision Tree

A Decision Tree is selected because the problem is a classification task with numerical and categorical input attributes.

The model recursively divides the data based on the most suitable attribute and produces a final decision of Approved or Rejected.

Attribute Selection Measure: Gini Index

The Gini Index is used to measure the impurity of a dataset.

$$\text{Gini Index} = 1 - \sum (p_i)^2$$

where p_i is the probability of a sample belonging to class i .

A lower Gini value represents a purer split.

Justification:

- Suitable for classification.
- Easy to interpret.
- Handles numerical and categorical attributes.
- Produces understandable decision rules.

6. MODEL DESIGN / ARCHITECTURE

Input Features

- Income
- Credit Score
- Employment Status
- Loan Amount
- Repayment History

Output

Loan Status: Approved / Rejected

7. ALGORITHM / PSEUDOCODE

Algorithm

1. Load the loan dataset.
2. Select the required input features.
3. Select Loan Status as the target.
4. Handle missing values.
5. Encode categorical attributes.
6. Split the dataset into training and testing sets.
7. Create a Decision Tree using the Gini criterion.
8. Train the model using training data.
9. Predict loan status for test data.
10. Evaluate the predictions using classification metrics.
11. Display the final results.

Pseudocode

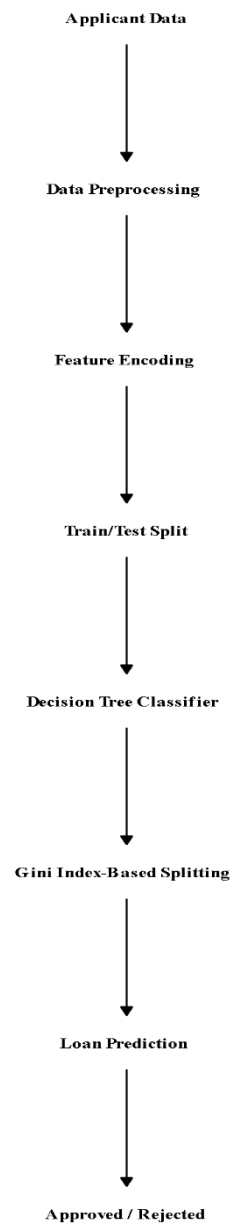
BEGIN

Load dataset

Select input features

Select Loan Status as target

Loan Prediction Process Flow



Preprocess the data
Encode categorical values
Split data into training and testing sets
Create Decision Tree using Gini Index
Train the model
Predict loan status
Calculate evaluation metrics
Display results
END

IMPLEMENTATION

The model is implemented using **Python**.

Libraries Used

- Pandas – dataset handling
- NumPy – numerical operations
- Scikit-learn – Decision Tree and evaluation
- Matplotlib – visualization

Implementation Steps

1. Import the required libraries.
2. Load the dataset.
3. Preprocess the data.
4. Encode categorical features.
5. Split the data into training and testing sets.
6. Train the Decision Tree classifier.
7. Predict loan status.
8. Calculate performance metrics.
9. Generate the confusion matrix and Decision Tree visualization.

EXPERIMENTAL RESULTS

The trained model is tested using unseen test data.

Sample Test Cases

	Case	Credit Score	Income	Loan Amount	Repayment History	Expected Result
1		High	High	Moderate	Good	Approved
2		Low	Low	High	Poor	Rejected
3		High	Moderate	Moderate	Good	Approved
4		Low	Moderate	High	Poor	Rejected

10. PERFORMANCE EVALUATION

The following metrics are used to evaluate the classification model:

Accuracy

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

Measures the percentage of correctly classified applications.

Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Measures the correctness of positive predictions.

Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Measures the ability to identify actual positive cases.

F1-Score

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Measures the balance between precision and recall.

Confusion Matrix

The confusion matrix contains:

- True Positive
- True Negative
- False Positive
- False Negative

11. RESULT ANALYSIS AND INTERPRETATION

The Decision Tree classifies applicants into **Approved** and **Rejected** categories based on their input features.

The model's performance is determined from the actual test results and evaluation metrics. Features such as credit score, income and repayment history can provide important information for classification.

The confusion matrix helps identify correct and incorrect predictions, while accuracy, precision, recall and F1-score provide quantitative measures of model performance.

The final analysis should be based on the actual results obtained from the implemented model.

12. LIMITATIONS

1. Model performance depends on the quality and size of the dataset.
2. A Decision Tree may overfit if its depth is not controlled.
3. Historical data may contain bias.
4. The model can only use the information provided in the dataset.
5. Real-world loan approval requires additional financial and regulatory considerations.

13. FUTURE ENHANCEMENTS

1. Use a larger and more diverse dataset.
2. Apply hyperparameter tuning.
3. Compare Decision Tree with Random Forest and other classifiers.
4. Add additional financial features.
5. Develop a web-based prediction application.
6. Add explainability for individual predictions.
7. Retrain the model using updated data.

14. CONCLUSION

The proposed Smart Loan Approval system uses a **Decision Tree classification model** to predict loan eligibility from applicant information. The system performs data preprocessing, model training, prediction and performance evaluation.

The Decision Tree provides an interpretable approach for classifying applicants as **Approved** or **Rejected**. Its performance is evaluated using appropriate classification metrics and test results.

Structure of the Report:

S.No.	Report Component	What the Student Should Include
1	Problem Statement	Clearly describe the real-world problem, input data, expected output, and AI task to be solved.
2	Objectives	State the specific goals of developing the AI model and expected outcomes.
3	Dataset Description	Provide dataset source, number of instances, features, target variable, data types, and important attributes.
4	Data Preprocessing	Explain data cleaning, missing-value handling, encoding, normalization/scaling, feature selection, and train-test splitting.
5	Selected AI Technique and Justification	Identify the selected AI technique and justify its suitability for the problem.
6	Model Design / Architecture	Present the model structure, components, input/output, parameters, hyperparameters, and architecture/flow diagram.
7	Algorithm / Pseudocode	Provide the algorithmic steps or pseudocode showing model training and prediction/decision-making.
8	Implementation	Provide programming language, libraries/tools, important code sections, and implementation details.
9	Experimental Results	Present results using tables, graphs, sample predictions, learning curves, confusion matrices, or other relevant outputs.
10	Performance Evaluation	Evaluate the model using appropriate performance metrics such as accuracy, precision, recall, F1-score, MSE, RMSE, or cumulative reward.
11	Result Analysis	Interpret results, identify important patterns, compare actual and predicted outcomes, and discuss strengths and weaknesses.
12	Limitations	Discuss limitations related to dataset, model complexity, computational resources, accuracy, generalization, or deployment.
13	Future Enhancements	Suggest improvements such as additional data, feature engineering, hyperparameter tuning, alternative algorithms, or deployment.
14	Conclusion	Summarize the problem, methodology, major findings, model performance, and overall effectiveness.
15	References	List datasets, research papers, books, websites, documentation, libraries, and other sources used.

16	GitHub Repository Link	Provide the public GitHub repository URL containing the complete source code, dataset information/source, implementation, results, README, and supporting files.
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Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Problem Analysis and Understanding	10%	9–10% – Clearly analyzes the real-world problem, objectives, inputs, outputs, constraints, and expected AI outcome.	7–8% – Understands the problem and objectives with minor omissions.	5–6% – Shows partial understanding with some important elements missing.	0–4% – Problem is poorly understood or incorrectly defined.
2. Dataset Selection and Understanding	10%	9–10% – Selects a highly relevant dataset and clearly explains features, target, source, size, and characteristics.	7–8% – Appropriate dataset with adequate description and minor omissions.	5–6% – Dataset is usable but description or relevance is limited.	0–4% – Dataset is inappropriate, poorly described, or missing.
3. Data Preprocessing and Feature Engineering	10%	9–10% – Performs appropriate cleaning, transformation, encoding, scaling, feature selection, and data splitting with clear justification.	7–8% – Performs most required preprocessing correctly with minor issues.	5–6% – Basic preprocessing is performed but important steps are missing.	0–4% – Preprocessing is inadequate, incorrect, or absent.
4. AI Technique Selection and Justification	10%	9–10% – Selects the most appropriate AI technique and provides strong technical justification based on the problem and dataset.	7–8% – Appropriate technique selected with reasonable justification.	5–6% – Technique is acceptable but justification is limited.	0–4% – Inappropriate technique or no meaningful justification.
5. Model Design / Architecture	10%	9–10% – Develops a clear, technically sound model architecture with appropriate components, parameters, and workflow.	7–8% – Model is well designed with minor omissions.	5–6% – Basic model design is presented but lacks technical depth.	0–4% – Model design is unclear, incomplete, or inappropriate.
6. Algorithm / Pseudocode	5%	5% – Algorithm/pseudocode is complete, logically correct, structured, and clearly represents the model development process.	4% – Mostly correct with minor omissions.	2–3% – Basic algorithm provided but contains gaps.	0–1% – Algorithm is missing or substantially incorrect.
7. Implementation and GitHub Repository	15%	13–15% – Fully functional implementation with clean, organized, reproducible code and a complete GitHub repository with README and supporting files.	10–12% – Functional implementation and well-organized repository with minor issues.	7–9% – Partially functional implementation or incomplete repository documentation.	0–6% – Implementation is incomplete/non-functional or GitHub repository is missing.

8. Experimental Design and Results	10%	9–10% – Conducts meaningful experiments with appropriate test cases and clearly presents results using tables, graphs, or visualizations.	7–8% – Appropriate experiments with clearly presented results.	5–6% – Limited experiments or basic result presentation.	0–4% – Insufficient, incorrect, or missing experimental results.
9. Performance Evaluation	10%	9–10% – Uses appropriate evaluation metrics and provides comprehensive quantitative analysis of model performance.	7–8% – Uses suitable metrics and provides adequate evaluation.	5–6% – Basic evaluation with limited metrics or analysis.	0–4% – Inappropriate metrics or performance evaluation is missing.
10. Result Analysis and Interpretation	5%	5% – Provides insightful analysis of results, identifies patterns, errors, strengths, weaknesses, and explains model behavior.	4% – Provides good interpretation with minor gaps.	2–3% – Basic interpretation with limited analysis.	0–1% – Results are not meaningfully analyzed.
11. Limitations and Future Enhancements	5%	5% – Clearly identifies realistic limitations and proposes technically relevant and feasible improvements.	4% – Identifies relevant limitations and reasonable improvements.	2–3% – Provides basic limitations and future suggestions.	0–1% – Missing, unrealistic, or poorly explained.
12. Documentation and Technical Presentation	10%	9–10% – Report is comprehensive, well organized, technically accurate, professionally presented, and properly referenced.	7–8% – Well documented with minor presentation or referencing issues.	5–6% – Adequate documentation with several gaps.	0–4% – Poorly documented, unclear, or incomplete report.
Total	100%				